

Active 3D Mapping Leveraging Heterogeneous Crowd Robot based-on Reinforcement Learning

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ABSTRACT

The paper demonstrates that the introduction of heterogeneous crowd robots improves the accuracy and completeness of indoor 3D mapping. First, three ground robots (sweeping robot, inspection robot, and guidance robot) are used to perform active exploration and mapping tasks in the iGibson indoor environment. However, the mapping results of the ground robot reveal that there are obvious blind spots in key areas (such as tables, stoves, and beds), resulting in the lack of point cloud data. To overcome this challenge, we introduce a drone for active 3D mapping using its bird's eye view and powerful perception capabilities. Experimental results show that by introducing drones, we have successfully eliminated the blind areas of vision existing in-ground robot mapping and achieved more comprehensive and accurate mapping results. This demonstration fully demonstrates the advantages of introducing heterogeneous crowd robots, and how the complementary capabilities of different types of robots can work together to improve the indoor 3D mapping process.

CCS CONCEPTS

• **Computer systems organization** → *Robotic autonomy*; • **Hardware** → *Sensor applications and deployments; Sensor devices and*

platforms; • **Human-centered computing** → *Ubiquitous computing*.

KEYWORDS

Crowd Robot, Ubiquitous Hardware, Reinforcement Learning, 3D Mapping

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1 INTRODUCTION

Indoor 3D mapping is widely applied in many fields such as digital twin, decoration, BIM, and education [7]. It plays a crucial role in interior design and renovation, enabling designers and contractors to visualize and plan indoor spaces, including residential homes, commercial offices, and hotels. Additionally, it contributes to digital twin experiences, creating immersive virtual environments for users to explore and interact with. Traditionally, indoor mapping is built by people with CAD files. This method is inefficient and with low accuracy due to the environment is always changed with the development of the building, and it will cause errors if you apply an indoor localization application in a digital twin system. To prevent this error, some people may use 3D laser scan equipment such as Trimble X7 [1] etc. However, this equipment is very expensive, and it needs well-trained people to operate and move it manually

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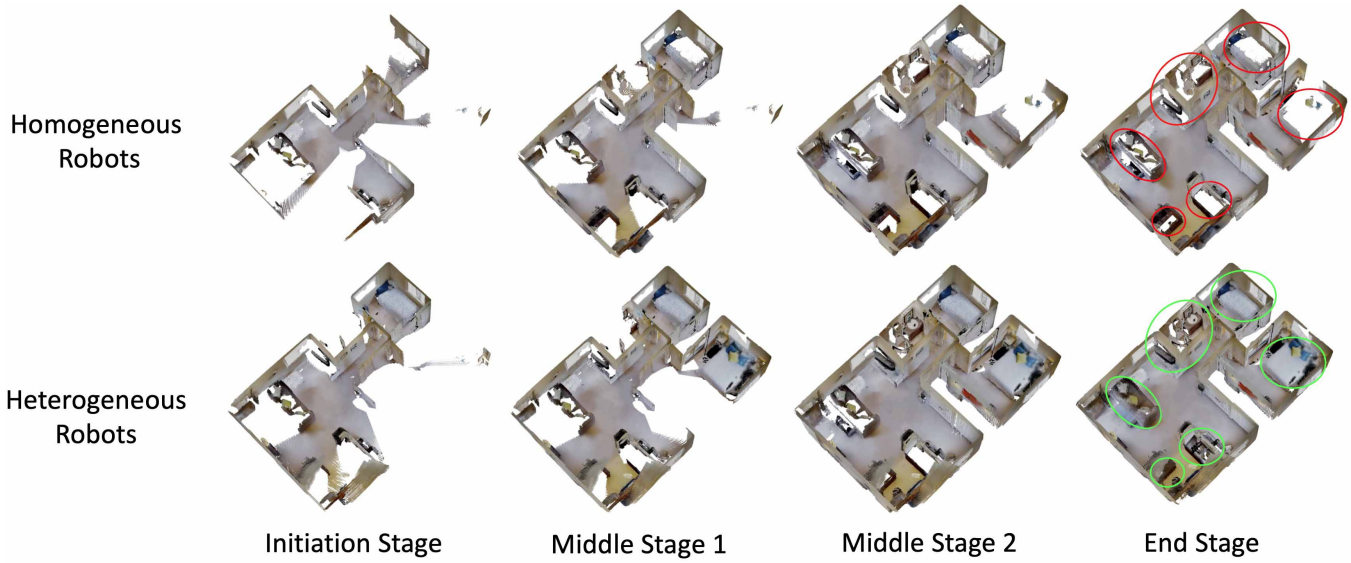


Figure 1: Examples of 3D mapping using homogeneous and heterogeneous robots

in the indoor environment. Also, this method is not able to cover all surfaces due to it needs to be set up on the ground and it has limitations in height. Thus you will find the missing point surfaces. In addition, Wang et al. [6] propose a multi-sensory backpack system for 3D mapping, but still it needs well-trained people to operate. Therefore we aim to develop an efficient crowd robot system to do 3D mapping.

Fortunately, with the rapid development of crowd sensing [2, 3] and robot applications, it's possible to combine these robots to do 3D mapping by using their RGBD sensors, etc. In this case, we propose a crowd robot system using heterogeneous robots, and the challenge is to develop an efficient and complicated 3D mapping system.

Therefore, we propose a crowd robot system. Initially, we employ an efficient crowd-sourced mapping technique to generate a basic map [4, 8] leveraging the contributions of homogeneous ground robots and reinforcement learning algorithms, such as cleaning robots, inspection robots, or guide robots. However, this approach may encounter limitations due to the height of the sensor and it won't cover the surface like the desktop etc. To overcome these limitations, we incorporate drones with RGBD sensors. By leveraging continuous boundary detection algorithms, we employ drones to reconstruct detailed structures by 3D route planners. This approach

allows us to capture intricate details of the environment, ensuring comprehensive and accurate 3D mapping.

Finally, to demonstrate the effectiveness and completeness of our method, we have developed a demo in iGibson [5]. In this scenario, we created an experimental environment and designed multi-robots. This showcases the potential of our approach in facilitating the efficient and complete mapping of indoor spaces, paving the way for applications in fields such as architecture, interior design, facility management, and virtual simulation.

2 CROWD ROBOTS SYSTEM

We have developed a crowd robot system that aims to achieve an efficient and complete 3D mapping process. This system leverages the Robot Operating System (ROS) and the iGibson simulator to control the robots and virtualize the resulting point cloud data from the 3D mapping process. Fortunately, the majority of heterogeneous robots in our system are compatible with the ROS framework.

By utilizing ROS, we establish a unified platform for communication and coordination among the heterogeneous robots. This allows us to seamlessly integrate different robot types, such as cleaning robots, inspection robots, and guidance robots, into a cohesive system. The robots can exchange information, share data, and collaborate effectively in order to accomplish the task of 3D mapping.

Furthermore, the integration of the iGibson simulator provides a realistic virtual environment for simulating and visualizing the point cloud results of the 3D mapping process. This virtualization allows us to evaluate and refine the mapping algorithms and strategies in a controlled and reproducible manner.

The compatibility of heterogeneous robots with the ROS system greatly simplifies the implementation and deployment of our crowd robots system. It enables us to leverage the existing capabilities and functionalities of these robots, while also facilitating seamless communication and interoperability between them. This compatibility

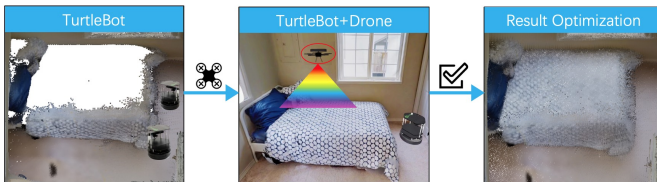


Figure 2: An example of a drone using an RGBD camera to capture the blind spot of a ground robot's field of vision

ensures that the system can effectively utilize the unique strengths and features of each robot type to achieve an efficient and complete 3D mapping system.

3 DEMONSTRATION

In the demonstration, we aim to showcase the improved accuracy and completeness of 3D mapping by introducing heterogeneous crowd robots. Firstly, we utilize three ground robots (a sweeping robot, an inspection robot, and a guidance robot) to actively explore and map the indoor environment in iGibson. During the mapping process, we employ reinforcement learning to maximize long-term values, thus enhancing both time efficiency and map completeness. However, the mapping results from the ground robots (indicated by the red circle in Figure 1) reveal noticeable blind spots in critical areas such as tables, stovetops, and bed surfaces, leading to missing point cloud data. To overcome this challenge, we introduce a drone, leveraging its aerial perspective (as shown in Figure 2) and powerful perception capabilities, for active 3D mapping. The results demonstrate that by incorporating the drone, we successfully eliminate the blind spots (indicated by the green circle in Figure 1) present in the mapping performed by the ground robots, thereby achieving more comprehensive and accurate mapping results. Additionally, our heterogeneous crowd robots system is highly scalable, allowing for the addition of specified numbers of robots to meet efficient mapping requirements in various indoor environments.

4 FUTURE WORK

This demonstration fully showcases the advantages of integrating heterogeneous crowd robots and how the complementary capabilities of different robot types can collaborate to improve the process of indoor 3D mapping. Such research is of significant importance in advancing the application of robotics technology in indoor environments, providing us with deeper insights into methods and tools for understanding indoor spaces and achieving more accurate environmental modeling capabilities.

In the future, we aim to introduce Quadruped robots to adapt to more complex mapping scenarios, leveraging the full potential of heterogeneous robots. Additionally, we will focus on developing advanced algorithms to effectively exclude the influence of dynamic obstacles, thereby enhancing the accuracy and stability of the mapping process.

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